Integration by parts (7.1)



The product rule for derivatives says

$$\frac{d}{dx}\left(f(x)g(x)\right) = f(x)g'(x) + f'(x)g(x)$$

Integration by parts: For any functions $f(x)$, $g(x)$

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\,dx$$

$$\int x\cos x\,dx$$

Integration by parts: For any functions $f(x)$, $g(x)$

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\,dx$$

Integration by parts: For functions $u = f(x)$ and $v = g(x)$,

$$\int u\,dv = uv - \int v\,du$$

$$\int x\cos x\,dx$$

$$\int x e^{2x} \, dx$$

$$\int \ln x \, dx$$

Integration by parts: For functions $u = f(x)$ and $v = g(x)$, defined on $[a, b]$,

$$\int_a^b u\,dv = uv\Big|_a^b - \int_a^b v\,du$$

$$\int_1^e x \ln x \, dx$$

$$\int_0^1 x^2 e^x \, dx$$

$$\int e^x \sin x \, dx$$

$$\int_0^1 \tan^{-1} x \, dx$$